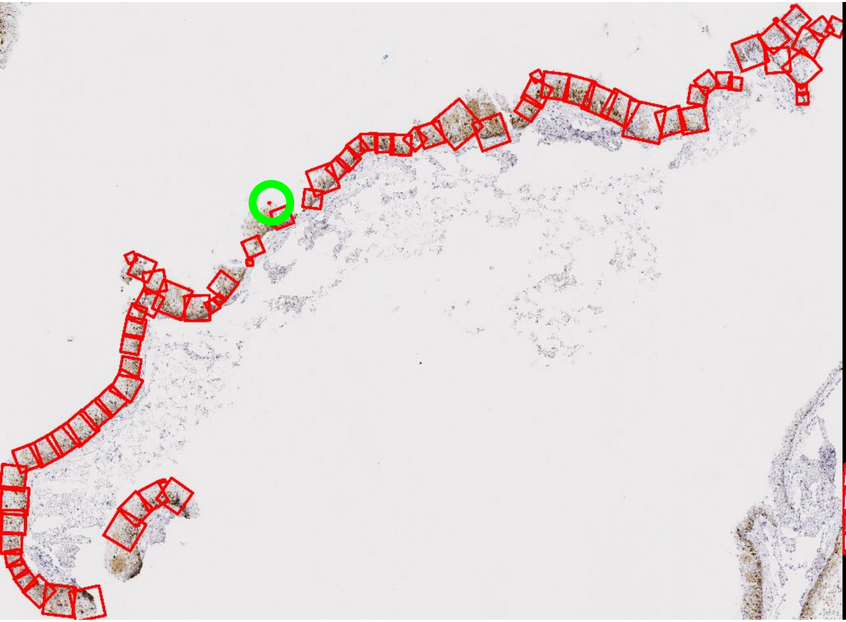
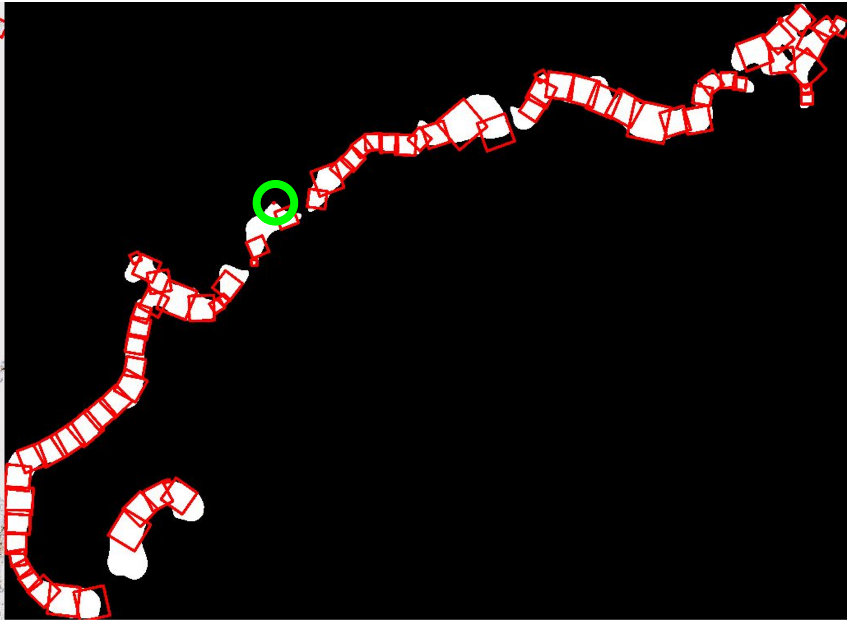
Small patch

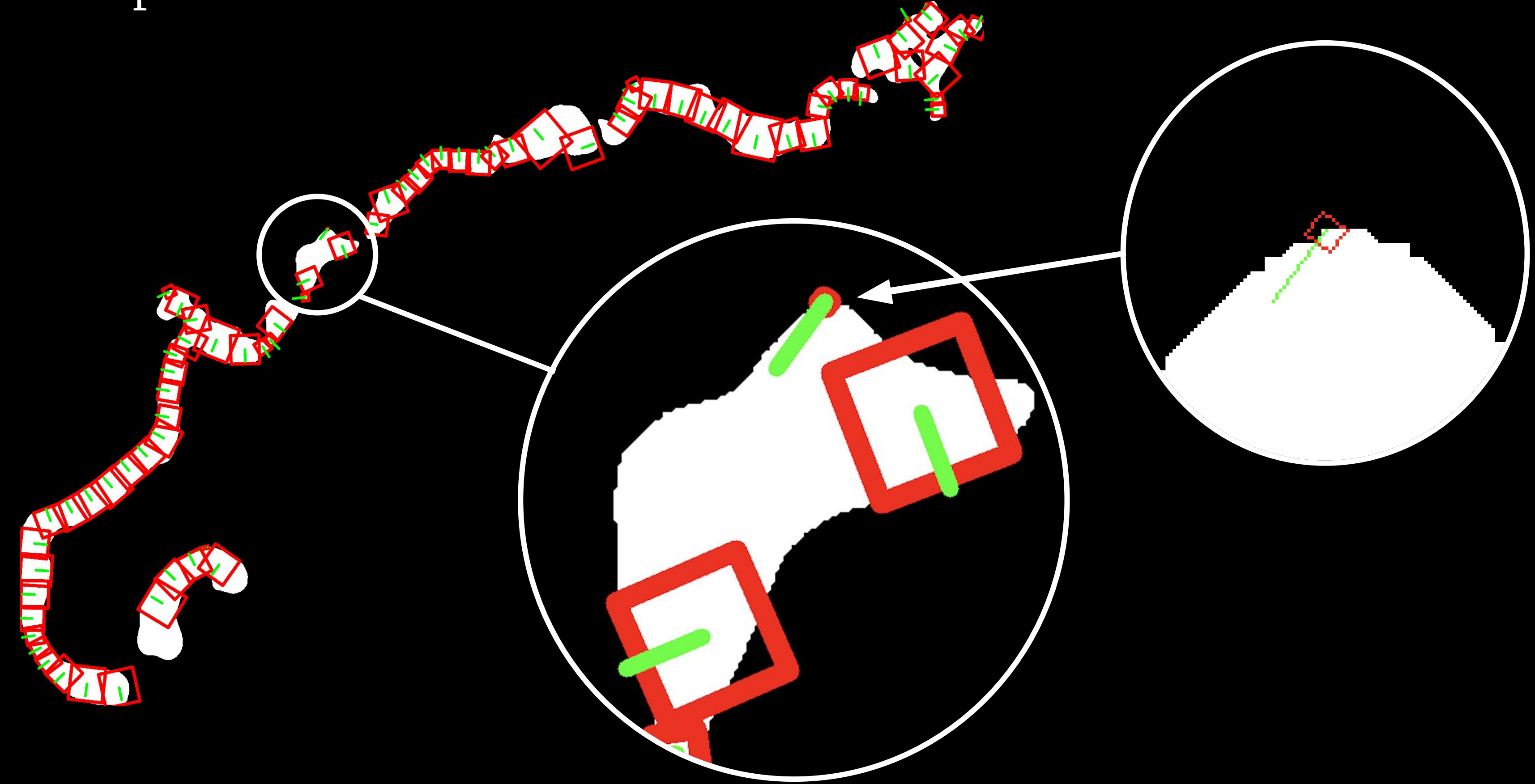




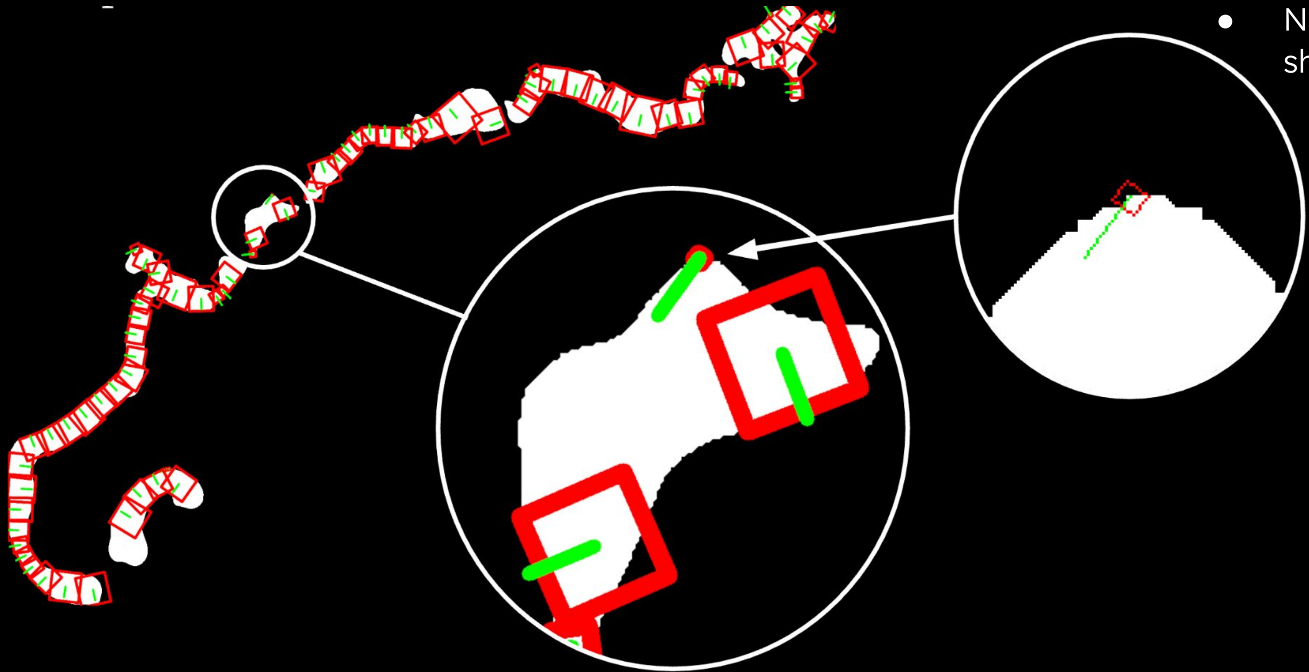
william wang

H,21.14175 Case 022 Melan

Small patch - first look

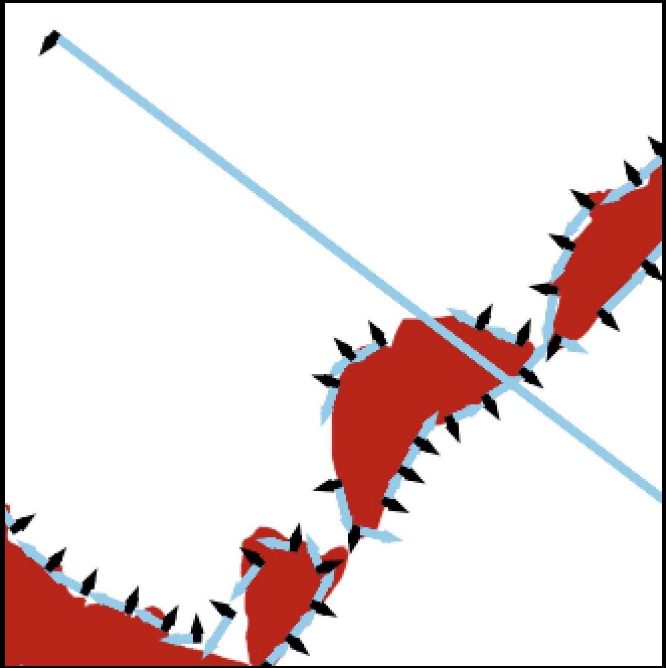


Small patch - first look



- Normal is drawn at an apex of the shape
  - Is something wrong with the tangent? What about the mask?
  - Is something wrong with how squares overlap?

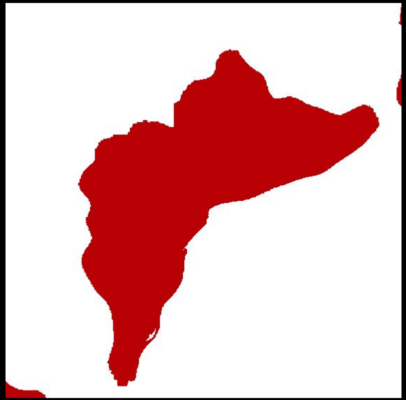
Small patch – tangent



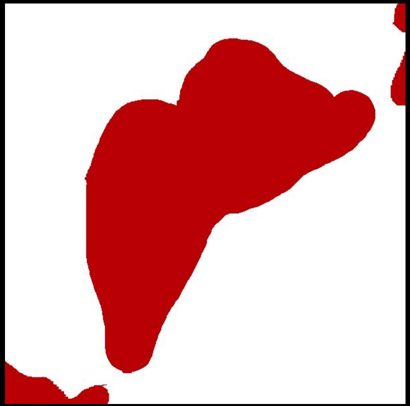
- Something is wrong with the tangent
  - Investigating noise and sharp edges
    - Masks
    - Tangent function

Masks

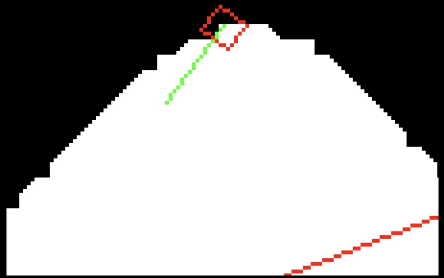
before



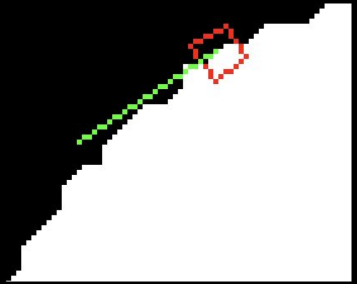
after



before



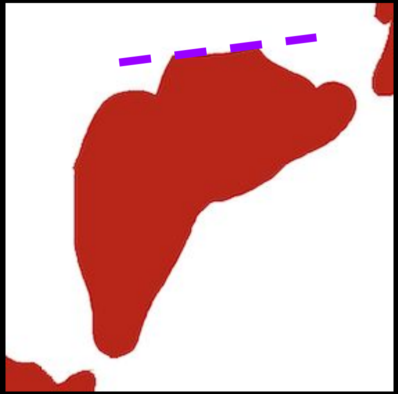
after



Is something wrong with the mask?

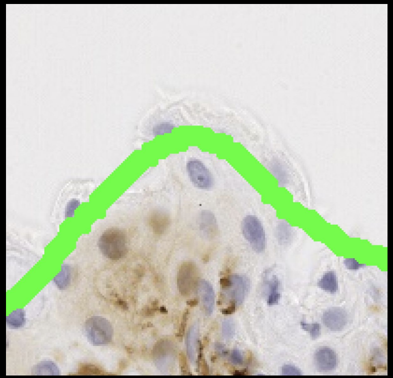
- Original mask was pretty jagged
  - Annotation is not the issue
  - Is it simply not smooth enough?

Masks



What if the annotation didn't facilitate in smoothing?

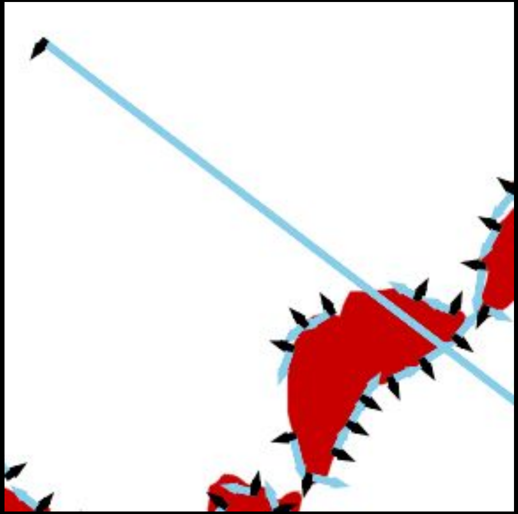
It doesn't matter



Contour seems fine too

Overlap threshold - first point fix

first point
np.gradient

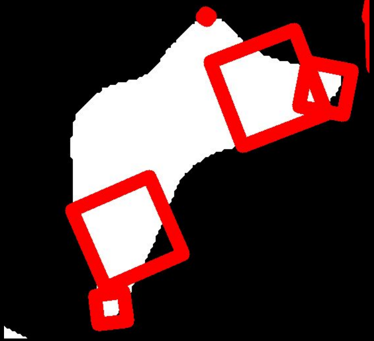


first point
np.diff

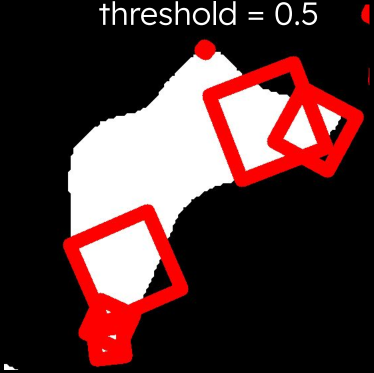
- Noise and Sharp Bends aren't too cohesive with np.gradient
- np.diff allows to set the first coordinate per contour with prepending.
- caveats for ALL 'odd' tangents

Overlap threshold

threshold = 0.3



threshold = 0.5

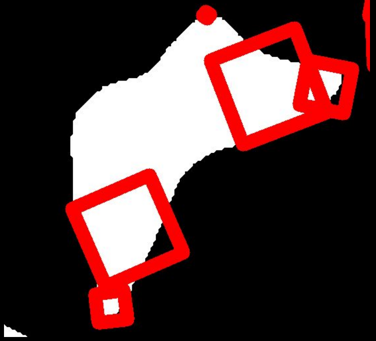


threshold = 0.9

Is something wrong with how squares overlap?

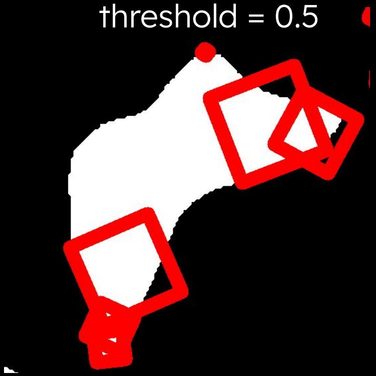
Overlap threshold

threshold = 0.3



threshold = 0.9

threshold = 0.5



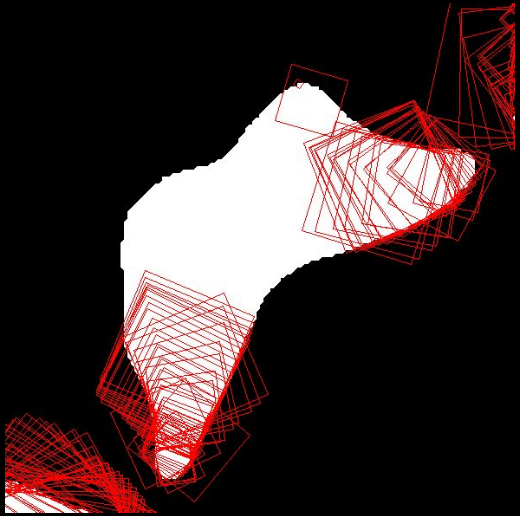
Is something wrong with how squares overlap?

Overlap threshold

Is something wrong with how squares overlap?

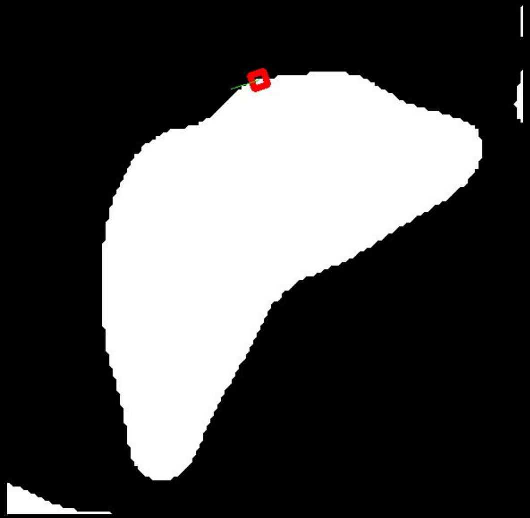
- Something's being negated regardless of threshold
    - The mask isn't the problem
- At threshold =1
    - All possible patches
    - Overlapping over smaller patch
- How are patches removed? What is the priority?

threshold = 1

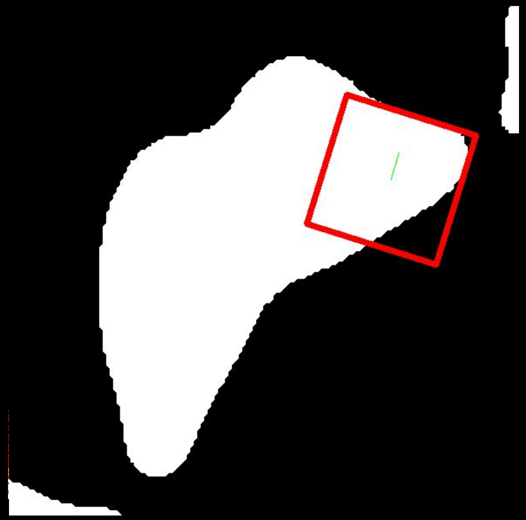


Overlap threshold – ‘priority’

threshold = 0.3
first patch

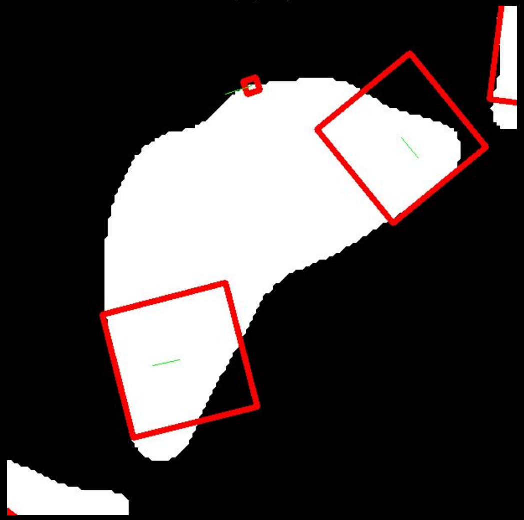


threshold = 0.3
last patch

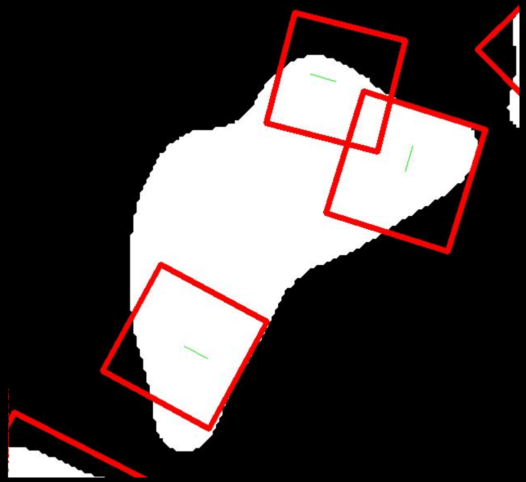


Overlap threshold – 'priority'

threshold = 0.3
as is

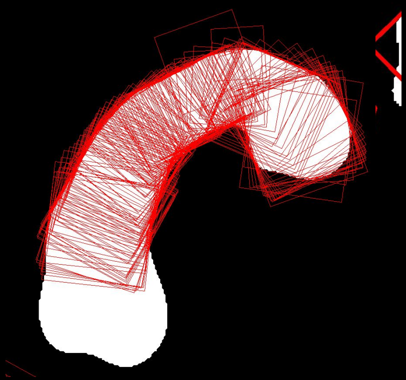


threshold = 0.3
reverse-order

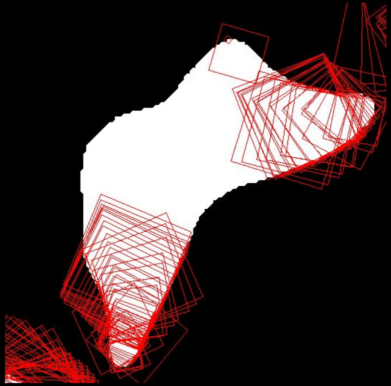


Overlap threshold - area tolerances

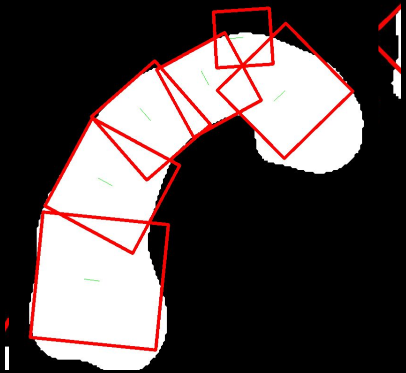
2-50%



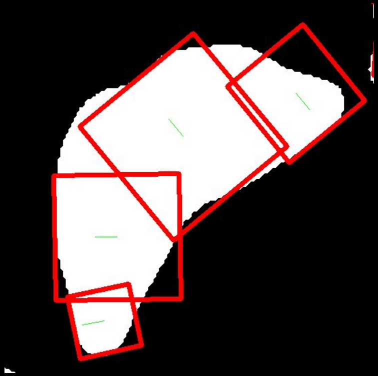
2-50%



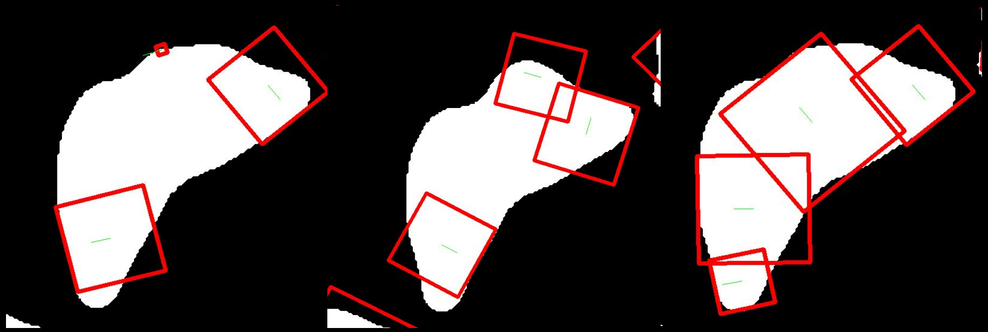
5-70%



5-70%



Takeaways



1. Sequencing Patch Removal
   a. Only reverse patch generation was demonstrated
   b. Masking is not the issue for once
2. Consider prioritizing by Patch size
   a. Brute forcing by creating size constraints is arbitrary and unindeal
3. Improving first patch logic and tangent logic for first patch
   a. Indirect alternative to brute forcing patch size